## CALC-01 — Limits

A **limit** describes the behavior of a function as its argument approaches a certain value.

$$\lim_{x \to a} f(x) = L$$

indicates that as $x$ approaches $a$, $f(x)$ approaches $L$. → *CALC-02 Derivatives* → *CALC-03 Continuity*

#limits #analysis #functions

## CALC-02 — Derivatives

The **derivative** of a function measures the rate of change of the function with respect to its variable.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

represents the slope of the tangent line at point $x$. → *CALC-01 Limits* → *CALC-04 Applications*

#derivatives #rateofchange #tangents

## CALC-03 — Integrals

An **integral** represents the accumulation of quantities, often interpreted as the area under a curve.

$$\int_a^b f(x)\,dx$$

calculates the total area between the function $f(x)$ and the x-axis from $a$ to $b$. → *CALC-02 Derivatives* → *CALC-05 Fundamental Theorem*

#integrals #area #accumulation

## CALC-04 — Fundamental Theorem of Calculus

The **Fundamental Theorem of Calculus** links differentiation and integration, stating: If $F$ is an antiderivative of $f$, then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

→ *CALC-03 Integrals* → *CALC-06 Applications*

#fundamentaltheorem #calculus #connections

## CALC-05 — Applications of Derivatives

Derivatives are used to find **critical points**, determine **increasing/decreasing** intervals, and analyze **concavity**.

$$f'(x) = 0$$

identifies critical points for optimization. → *CALC-02 Derivatives* → *CALC-06 Applications*

#applications #optimization #criticalpoints

## CALC-06 — Applications of Integrals

Integrals are applied in calculating areas, volumes, and solving differential equations. For example, the area between curves is given by:

$$A = \int_a^b (f(x) - g(x))\, dx$$

→ *CALC-03 Integrals* → *CALC-04 Fundamental Theorem*

#applications #area #volumes

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